

A hybrid deep learning approach for efficient detection and classification of internal eye diseases

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ABSTRACT

Glaucoma is a progressive optic neuropathy and one of the most common causes of the irreversible blindness in the world. Traditional diagnostic tools like Optical Coherence Tomography (OCT) and visual field testing are also dependable but usually costly, laborious, and not accessible in low-resource healthcare settings. Also, the interpretation of fundus images manually causes inter-observer variations and delays in diagnosis. Despite some promising results in the field of deep learning applications in the detection of glaucoma, the current models are often limited due to overfitting and low ability to generalize as well as poor temporal progression. This research suggests an innovation to overcome all these challenges by proposing a hybrid deep learning architecture that combines spatial feature extraction of convolutional neural network (CNN) with handcrafted descriptors such as Local Binary Patterns (LBP), Histogram of Oriented Gradients (HOG), and Speeded-Up Robust Features (SURF). The temporal dynamics are shaped with the help of the Gated Recurrent Unit (GRU), and the fusion of the features is optimized with the help of the Maximum Relevance Minimum Redundancy (mRMR) approach to decrease the redundancy and maximize the discriminative power. The classification is done with various machine learning classifiers which are Multilayer Perceptron (MLP), GRU, Support Vector Machine (SVM), Random Forest (RF), and k-Nearest Neighbors (KNN) to maintain robustness and interpretability. The framework is tested using publicly available datasets of glaucoma fundus images, such as ORIGA, REFUGE and G1020, after being preprocessed (resizing, normalization, region of interest (ROI) localization, and segmentation using U-Net). Experimental findings show better performance with 99.7% accuracy, 99.6% precision, 99.5% recall and 99.55% F1-score, which is better than the state of art methods. The proposed system is a scalable, cost-effective, and clinically relevant system implemented in Python with the help of TensorFlow, OpenCV, and Scikit-learn, to detect glaucoma in its early stages and monitor progression.

1. Introduction

Glaucoma is an optic neuropathy that develops progressively and is one of the most important causes of blindness that cannot be reversed in the world (Jalalova et al., 2025). It is mainly due to the elevated intraocular pressure which destroys the head of the optic nerve and causes gradual loss of vision in case it goes untreated (Sharif, 2022). The cup and optic disc areas of the retina are considered to be the most important parts of the eye in the assessment of glaucoma and the morphological alterations of these areas are an essential tool in the diagnosis of the disease (Veena et al., 2022). A sample fundus image of a normal eye and a glaucomatous eye is given in Fig. 1. The WHO postulates that currently more than 76 million people worldwide have

glaucoma, and this figure is projected to increase to 111.8 million in 2040 (Godinez, 2024), (Akbar et al., 2024). Early screening and correct classification are profoundly important in preventing blindness since the symptoms tend to manifest in late stages. Fundus photography has become an important method of diagnosis; it is a diagnostic technique that allows visualizing a high-resolution picture of the retina and diagnosing glaucoma (Ali et al., 2024).

Traditional diagnostic procedures like OCT, Visual Field Testing and Gonioscopy are common clinical diagnosis tools (Alexopoulos et al., 2022). The interpretation of these modalities is however, often tedious, time consuming and prone to inter-observer variability with manual interpretation. Besides, these diagnostic systems are not only costly, but also inaccessible to some healthcare organizations, especially

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low-resource institutions (Olatunji et al., 2024), (Ramachandran et al., 2022). ML and DL are automated systems, which have become possible solutions to overcome these shortcomings. Especially CNNs, have shown potential performance in retina fundus images analysis and high-level features of optic disc and cup morphology (Alfonso-Francia et al., 2022). Although they have shown to be effective, current DL methods cannot cope with issues, which include small dataset sizes, over-fitting, and a lack of temporal course of glaucoma, as well as poor generalization to diverse imaging environments (Shyamalee et al., 2024).

In order to address these constraints, this paper will suggest a Hybrid Deep Learning Framework to detect and classify glaucoma early and with high accuracy. The method combines CNN-based deep feature extraction with handcrafted features like LBP, HOG and SURF to extract both deep hierarchical and fine-grained textual features. To narrow-down the extracted features, a Maximum Relevance-Minimum Redundancy (MR-MR) feature selection procedure is incorporated so that the features are relevant and not redundant. To classify, a mixture of MLP, GRU and classical ML classifiers (SVM, Random Forest, KNN) is used in order to guarantee sound performance. The framework effectively incorporates spatial, sequential and handcrafted characteristics, which not only increase the accuracy of the diagnosis but also allow tracing the development of glaucoma throughout the years.

The study demonstrates a systematic hybrid system of glaucoma detection that incorporates deep CNN features, engineered descriptors, as well as biomarkers that are clinically explainable, in one diagnostic system. In contrast to current methods, which either make use of end-to-end deep learning or only handcrafted features, the method presented in this paper systematically combines the two complementary representations and optimizes them in mRMR feature selection to prune redundant features and have better generalization. Secondly, the framework assesses various classification schemes, such as MLP, GRU, and traditional machine learning models, as well as end-to-end CNN base to provide fair and transparent analysis of performances. It is an optimized and integrated design that boosts the accuracy of the diagnosis and preserves clinical relevance and interpretability.

The key contribution of this study is.

- Development of a Hybrid CNN–Handcrafted Features–ML Classifier framework for glaucoma detection using fundus images.
- Integration of MR-MR feature selection to optimize the feature space and reduce redundancy.

- Utilization of multiple classifiers (MLP, GRU, SVM, RF, KNN) to evaluate performance and improve robustness.
- Focus on enhancing both diagnostic accuracy and disease progression analysis, supporting clinicians in reliable decision-making.

1.1. “Research motivation

Glaucoma is a progressive chronic eye disorder that has been at forefront of irreversible blindness in the world. Its early stages are usually asymptomatic and can therefore be diagnosed later when the vision damage has been widely inflicted. The conventional diagnostic tools are OCT and visual field testing, which effective, but costly, require manual interpretation, and are not available in low-resource areas. In addition, the current automated models typically have limitations on the data, overfitting, and diminished robustness when applied to a wide range of populations. These issues demonstrate the necessity of an effective, scalable, and automated system that will be able to identify glaucoma at the earliest stage, categorize stages of the disease accurately, and monitor their developmental trends. As the hybrid deep learning methods that integrate CNN-based feature extraction, handcrafted features, and state-of-the-art classifiers are introduced, proposed strategy is driven by the fact that the gaps in the accuracy, accessibility, and clinical applicability of large-scale glaucoma screening and diagnosis can be closed.

1.2. Research significance

The relevance of the research is that it can contribute to the enhancement of automated glaucoma detection using hybrid DL architecture based on the combination of various feature extraction and classification methods. Over traditional techniques that mainly use CNNs or manually designed features, this research paper integrates the use of spatial, textural, and temporal data to improve a diagnostic accuracy and generalizability. The MR-MR feature selection is the guarantee of computational efficiency, and the classifiers of MLP, GRU, SVM, RF, and KNN will give the comparative performance information. Medically, the framework provides the ophthalmologists with a decision-support tool that reduces human error, minimizes the time to diagnosis, and provides early intervention approaches to conserve the vision. In a bigger picture, the suggested model will help to provide scalable, cost-effective eye care solutions, thus glaucoma screening will

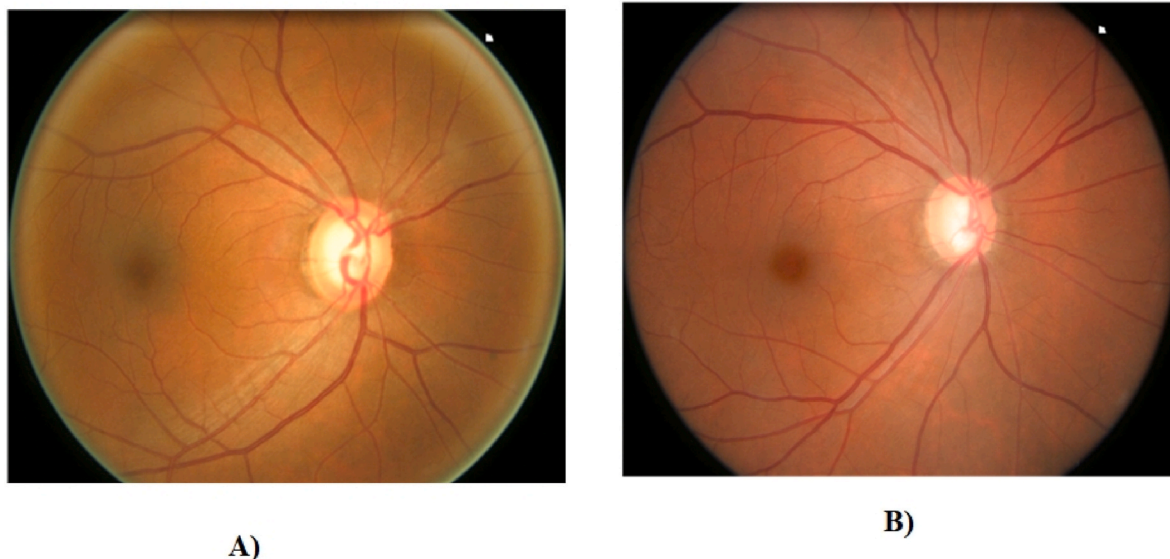


Fig. 1. Fundus Images showing (A) Healthy Eye (b) Glaucomatous Eye.

be more available in underprivileged areas. In the end, the study is important to fill the gap between the latest AI developments and their application in the real-life ophthalmology field."

1.3. "Problem statement

Glaucoma is a significant cause of permanent blindness globally, and it can develop without any warning up to an advanced level. Early diagnosis is critical but manual diagnosis with images of the retinal fundus is prone to inter-observer variance and time-consuming. Conventional machine learning techniques have a poor ability to reflect the fine structural and textural variations in the optic nerve head, particularly in different populations and imaging. Proper localization of the OC and OD is essential in determining clinical biomarkers like the CDR, which is an important measure of glaucomatous damage. Low contrast, vessel occlusions and anatomical variability all challenge segmentation, however. Current models tend to be uninterpretable and do not combine handcrafted and clinical characteristics with deep learning representations (Jauro et al., 2024). To overcome these, the study introduces hybrid deep learning architecture incorporating CNNs to extract spatial features, MLPs to classify diseases, and GRUs to analyze patterns over time. The model can be used to track the progression of the disease dynamically by fusing the characteristics of a static image with sequential patient data. This multi-architecture combination improves the accuracy and interpretability of diagnostic results, and provides a broad-scale solution of the problem of early detection and classification of glaucoma."

"The rest of the paper is structured in the following way: Section 2 is the discussion of some previous works and the related work. Section 3 gives the problem statement. Section 4 describes the methodology, such as preprocessing of data sets, feature extraction and classification. Section 5 presents the results and discussion, and Section 6 is a conclusion of the paper that includes the major findings and future perspectives."

2. "Literature review

Shamsan et al. (Shamsan et al., 2023) is to create an automated diagnostic method of early eye disease diagnosis through Colour Fundus Photography (CFP). These methods consist of three hybrid methods that are based on deep learning models (MobileNet and DenseNet121), PCA, and manual features to extract and fuse them followed by classification using an ANN. The data set involved is the CFP images that are identified by various eye diseases. The method has such benefits as it has better classification, dimensionality, and diagnostic performance. An AUC of 99.23 with accuracy of 98.5 was the best obtained with the combination of fused MobileNet and handcrafted features. The drawback is that fusion methods are complicated and require a particular dataset, which can influence the generalization.

Vardhan et al. (Vardhan et al., 2024) focus on creating an effective and precise system grounded in deep learning to detect the key diseases of the eye, such as Diabetic Retinopathy, Cataract, and Glaucoma, at an early stage so that relevant intervention can be performed to minimize the risk of losing sight. The research makes use of large datasets of medical images with labeled eye images to train and test the models. The proposed methodology will entail the use of popular DL models VGG19, InceptionV3 and ResNet50 along with the transfer learning approach to automatically identify disease-relevant features and enhance classification. The benefits of this method are that it will make the diagnosis quicker, more specific when detecting various eye diseases and it will also eliminate the aspect of manual expertise. The findings indicate that ResNet50 has the best accuracy of 99.94, precision, recall, and F1 of 96.28, 96.2, 96.24 respectively, which is better than VGG19 and InceptionV3. This shows that transfer learning is effective in boosting the performance of DL in the detection of ocular diseases. The disadvantages are that huge, excellent datasets are required and deep learning models are expensive to train.

ARSLAN and Erdaş (Arslan & Erdaş, 2023a) aims to create an efficient strategy that detects eye diseases including cataract, diabetic retinopathy, and glaucoma with help of DL to improve the diagnosis accuracy and patient care. The researchers used five Convolutional neural networks (CNN) models, which include: DenseNet, EfficientNet, Xception, VGG, and ResNet to classify retinal fundus images. This dataset has 2748 pictures of 1374 healthy and 1374 diseased people. The approach has such benefits as automatic detection of diseases and comparative analysis of CNN models. The highest performance was obtained when EfficientNet was used with an accuracy of 94.88, a recall of 94.88, a precision of 95.02, and a F1-score of 94.88. One of the limitations is that the model is reliant on the corresponding dataset which may restrict the extrapolation to other populations.

Nawaz et al. (Nawaz et al., 2022) create an automated deep learning model to identify early and accurate cases of glaucoma, eliminating the use of manual screening and the availability of experts. This is the method based on EfficientDet-D0 and EfficientNet-B0 as a backbone, deep feature extraction, BIFPN to combine the features, and lesion classification. The model was tested on the ORIGA data set, and verified against HRF and RIM-ONE DL data sets to guarantee the level of generalization. The major benefits are the enhancement of the quality of detecting complex glaucomatous areas and the minimization of delays during diagnosis. The model attained 0.97 recall and 0.979 AUC which is better than recent methods. The weakness is that there may be a variation in performance when there is variation in the imaging conditions or populations."

Kashyap et al. (Kashyap et al., 2022) is aimed at creating a precise glaucoma early detection system through deep learning, specifically to lower the rate of misdiagnosis that is high in underdeveloped medical environments. The technique is a composite of pretrained transfer learning models and U-Net in the task of OC segmentation and DenseNet-201 in the task of feature extraction and classification. The glaucoma data that was used in the study comprises retinal fundus images. Some of the main benefits are early symptomless detection, high accuracy, and better performance when compared to traditional CNN models. The model was shown to have 98.82 training accuracy with a training recall of 98.95, and a test accuracy of 96.90 and a test recall of 97.03. One of the weaknesses is that the performance of the model might differ in different populations and imaging situations.

Islam et al. (Islam et al., 2021) aims to create an automated, precise, and economical tool of glaucoma detection to eliminate problems that come with manual diagnosis. The researchers trained several DL models: EfficientNet, MobileNet, DenseNet, and GoogLeNet, on a novel personal dataset of 634 color fundus images labeled with expert ophthalmologists. Also, a dataset of blood vessel segmentation was produced with the help of a U-Net model that was trained on the High-Resolution Fundus Image Database. The main benefits are a decrease in the reliance on the use of costly tools and professionals, and the early identification of glaucoma. EfficientNet-b3 gave the best performance with the highest test accuracy of 0.9652, F1-score of 0.9512, and AUC of 0.9574. One of the weaknesses is the use of quite small, privately-held data, which could have implications on the ability to generalize to more general populations.

Arrieta-Rodriguez et al. (Arrieta-Rodriguez et al., 2025) aims to conduct a systematic review of the recent deep learning and computer vision methodologies in glaucoma detection by fundus image to detect major trends, techniques, and limitations. PRISMA methodology was used to review 63 publications published in 2020-24 on the databases of major databases. Approaches discussed are CNNs (and ResNet, in particular), U-Net, attention-guided networks, hybrid networks, and transformer networks, used in segmentation, feature extraction, and classification. Its strengths are high accuracy (up to 98.48) on binary and multiclass detection and particularly with attention mechanism and curriculum learning strategy. Findings demonstrate the efficiency of the hybrid and transformer-based models, and the accuracy can also be over 95%. Limitations are the absence of external validation in the vast

majority of studies, excessive dependence on structural features, and a difference in the metrics of evaluation between datasets.

Mathew et al. (Mathew et al., 2023) aimed at creating a workable means of detection of glaucoma by use of retinal fundus images (RFI). The research uses images of retina that contain demarcated images of optic discs and cups in glaucoma diagnosis. The preprocessing is done using a kernel bilateral filter (kernel BF) to increase image clarity and minimize noise and then brightness is altered by linear histogram transformation (LHT). Two-stage thresholds U-Net (Two-stage TUNet) is used to segment the optic disc and cup, and features are categorised with the help of the ensemble DeepNet comprised of VGGNet, DenseNet, and CapsNet via weighted majority voting, and the optimal weights were selected by the improved aquila optimization (IAO) algorithm. The proposed solution has the following benefits: precise early-stage identification, better image quality and a high level of classification with the help of ensemble learning. The model has a high accuracy of 99.45 %, which shows that the model is effective in predicting glaucoma. Possible drawbacks, however, are an additional level of computational effort as a result of using ensemble models and reliance on high-quality labeled data.

Joshi et al. (Joshi et al., 2022) is aimed at creating a computer-aided design (CAD) system to screen, diagnose and detect glaucoma at early stages by taking fundus images. The research employs 5 datasets of retinal fundus images including PSGIMSR, DRISHTI-GS, DRIONS-DB, HRF and mixed-dataset of retinal fundus images and optic discs and cups. The procedure includes an ensemble-based deep learning model based on combining three already trained convolutional neural networks: ResNet, VGGNet, and GoogLeNet to classify glaucoma. The benefits of the method are that they allow automated early detection, provide support to ophthalmologists with an effective second opinion, and analyze the severity of the disease based on segmented optic discs and cups structural characteristics. The proposed model had a high accuracy of 91.11, 95.63, 98.67, 95.64 and 88.96 on PSGIMSR and on DRIONS-DB, HRF, DRISHTI-GS and constellations, respectively. The model shows positive results with regard to early diagnosis of glaucoma and assessment of the condition of the eye. The drawbacks are however that it requires high-quality labeled datasets and may be computationally complex because it uses the ensemble deep learning model.

Zhang et al. (Zhang et al., 2024) aimed at creating an automatic

classification system of early and moderately advanced glaucoma so as to allow automatic grading and timely intervention. The paper relies on a privacy dataset of 2D fundus pictures and 3D-OCT scanner pictures to find useful characteristics that can be applied to their classification. The methodology utilizes a 2-branch deep learning network with EfficientNetB3 on 2D fundus images and ResNet34 on 3D-OCT images modified with an attention mechanism to highlight relevant features in the eye and discard useless background noise. The benefits of this method are enhanced feature extraction, less redundancy, classical grading of glaucoma stages and an ophthalmologist second opinion. The model had a high classification rate of 97.83% which shows its ability to screen glaucoma. The proposed system facilitates a rapid diagnosis, minimizes misdiagnosis, and helps to make a timely treatment decision. The possible drawbacks are that it will depend on a personal dataset since this will restrict generalization, and it will be computationally expensive because of two-branch attention architecture.

Table 1 reviewed underline that early glaucoma and eye disease detection based on fundus and OCT images are implemented by different approaches of deep learning. Hybrid frameworks, CNN variants, U-Net segmentation, and ensemble methods had high accuracies with many having greater than 95%. Although it has been demonstrated that the results can be used as a strong diagnostic tool, the limitation of the reliance on large, high-quality datasets, computational complexity, poor generalization across populations, and lack of external validation have been noted, which makes scalable, cost-effective, and robust clinical solutions a high necessity.

3. Hybrid CNN-MLP-GRU framework for d detection and classification of glaucoma eye diseases

This study presents a comprehensive methodology for glaucoma detection and classification using retinal fundus images. It starts with those data that are curated by collecting data in publicly available datasets and then processing those data to enhance the image quality and consistency which includes steps of resizing, normalization, noise removal, and region of interest (ROI) extraction. It utilizes a segmentation model to isolate essential anatomical structures to calculate clinically relevant bio-markers. The feature extraction technique integrates deep learning representations with handcrafted descriptors and

Table 1
Existing studies on automated eye disease and glaucoma detection using deep learning approaches.

Reference	Purpose of Study	Key Results	Disadvantages
Shamsan et al. (Shamsan et al., 2023)	Hybrid DL + PCA for early eye disease detection using CFP images.	Best AUC: 99.23%, Accuracy: 98.5% (fused MobileNet + handcrafted features)	Fusion methods are complex and require specific datasets, affecting generalization.
Vardhan et al. (Vardhan et al., 2024)	Detect key eye diseases early using DL (VGG19, InceptionV3, ResNet50) and transfer learning.	ResNet50 Accuracy: 99.94%, Precision: 96.28%, Recall: 96.2%, F1: 96.24	Requires large, high-quality datasets; deep learning models are expensive to train.
ARSLAN and Erdaş (Arslan & Erdaş, 2023a)	Efficient detection of cataract, diabetic retinopathy, glaucoma using CNN models (DenseNet, EfficientNet, Xception, VGG, ResNet).	EfficientNet Accuracy: 94.88%, Recall: 94.88%, Precision: 95.02%, F1: 94.88	Model dependent on the specific dataset; may limit generalization.
Nawaz et al. (Nawaz et al., 2022)	Automated glaucoma detection using EfficientDet-D0, EfficientNet-B0, BiFPN, and lesion classification.	Recall: 0.97, AUC: 0.979	Performance may vary with different imaging conditions or populations.
Kashyap et al. (Kashyap et al., 2022)	Early glaucoma detection using U-Net for optic cup segmentation and DenseNet-201 for feature extraction/classification.	Training Accuracy: 98.82%, Training Recall: 98.95%, Test Accuracy: 96.90%, Test Recall: 97.03	Performance may differ across populations and imaging conditions.
Islam et al. (Islam et al., 2021)	Automated, cost-effective glaucoma detection using DL models and U-Net for blood vessel segmentation.	EfficientNet-b3 Accuracy: 0.9652, F1-score: 0.9512, AUC: 0.9574	Small private dataset; may affect generalizability.
Arrieta-Rodriguez et al. (Arrieta-Rodriguez et al., 2025)	Systematic review of deep learning/computer vision for glaucoma detection on fundus images (CNNs, U-Net, attention, hybrid, transformer models).	Accuracy up to 98.48% (binary & multiclass); hybrid/transformer models >95%	Most studies lack external validation; excessive reliance on structural features; inconsistent evaluation metrics.
Mathew et al. (Mathew et al., 2023)	Early glaucoma detection using retinal fundus images, Two-stage TUNet for segmentation, ensemble DeepNet (VGGNet + DenseNet + CapsNet) for classification.	Accuracy: 99.45%	High computational effort; reliance on high-quality labeled data.
Joshi et al. (Joshi et al., 2022)	CAD system for ensemble deep learning-based early glaucoma detection (ResNet + VGGNet + GoogLeNet) on multiple retinal fundus datasets.	Accuracy: PSGIMSR: 91.11%, DRIONS-DB: 95.63%, HRF: 98.67%, DRISHTI-GS: 95.64%, Constellations: 88.96	Requires high-quality labeled datasets; computationally complex due to ensemble models.
Zhang et al. (Zhang et al., 2024)	Dual-branch DL (fundus + OCT) with attention for glaucoma grading.	Classification Accuracy: 97.83%	Personal dataset limits generalization; computationally expensive due to 2-branch attention architecture.

fuses and optimizes these two types of features with the idea of feature selection. Lastly, the diagnostic performance is assessed using multiple classifiers, which will be robust, interpretable, and have clinical relevance in glaucoma prediction. The visual representation is shown in Fig. 2.

The schematic depicts the entire workflow, comprising CNN-based deep features extraction, handcrafted features computation (LBP, HOG, SURF) as well as clinical biomarker extraction (CDR). The mRMR selection is used to obtain and fuse extracted features and further evaluate them on a variety of classifiers (MLP, GRU, SVM) to guarantee the robust and interpretable classification of glaucoma.

3.1. Data collection

The dataset used by the study is the Glaucoma Fundus Imaging Datasets ("Glaucoma Fundus Imaging Datasets") that consist of retinal fundus images and their OD/OC segmentations and glaucoma diagnosis labels. The datasets that have been incorporated in this study are ORIGA, REFUGE, and G1020 and they correspond to high-quality fundus images taken on different populations and at different imaging devices. The original images were then processed to produce cropped images with the optic disc at the center, and its respective OD and OC segmentation masks. The ORIGA dataset consists of a CSV file of image IDs and glaucoma labels (0; Healthy, 1; Glaucomatous), which can be used as the ground truth in classification tasks. Photos have been chosen to have an equal number of healthy and glaucomatous cases so that the training of models is strong. This selected dataset is the basis of designing the suggested hybrid DL model that allows the extraction of features, classifications, and temporal changes in glaucoma progression to be accurate. Fig. 3 illustrates the sample images from the dataset.

The research applies retinal fundus pictures of ORIGA and DRISHTI-GS1 data. A total of 750 images were utilized of which there were 218 health and 532 glaucoma images. An in-depth description of images per dataset and class is presented in Table 2. To train the model, 70% of the data was used in model training, 15% in model validation and 15% in model testing, with equal distribution of the classes of all the splits. To develop a model, the dataset was divided into three subsets consisting of 70% training, 15% validation, 15% testing to ensure powerful and unbiased model performance.

3.2. Data preprocessing

The first stage of a ML pipeline is data preprocessing in which raw data is cleansed, standardized, scaled and refined to be more quality and to be analyzed or trained into a model. This may encompass noise removal, contrast changes and extraction of region of interest in medical imaging.

3.2.1. Image resizing & standardization for CNN input

Each and every retinal fundus image was downsized to a fixed size (224 x 224) to make them consistent and compatible with CNN input.

Intensities of pixels were brought to a common range [0,1], or standardized through z-score normalization to have a mean of 0 and unit variance. This step of preprocessing assures that the CNN gets consistently scaled and contrasted images that allow it to effectively learn hierarchical details, such as OD and OC texture, vascular patterns, and other structural differences, which prove to be useful in identifying glaucomatous eyes by contrasting them with healthy ones.

3.2.2. Noise reduction (Gaussian filtering)

A method to minimize noise and smooth images while maintaining crucial structures like the OD and OC. It is expressed as in (1)

$$I_{G(x,y)} = \sum_{i=-k}^k \sum_{j=-k}^k I(x+i, y+j) \cdot G(i,j) \quad (1)$$

Where, $I(x, y)$ represent original image, $I_{G(x,y)}$ is filtered image, and $G(i,j)$ is the Gaussian kernel controlling the smoothing.

3.2.3. Region of interest (ROI) extraction

Isolation of most pertinent region of retinal image, the OD and OC is a process known as ROI extraction, which is aimed to concentrate categorization and feature extraction on the disease-relevant regions of the retinal image and eliminate irrelevant background. Mathematically, if $I(x, y)$ the original image is represented as $I(x, y)$ and the bounding box of the ROI to which the top-left corner belongs. (x_1, y_1) and bottom-right corner (x_2, y_2) , the cropped image I_{ROI} can be expressed in (2)

$$I_{ROI} = I[x_1 : x_2, y_1 : y_2] \quad (2)$$

Here, I_{ROI} only includes the pixels in the optic disc/cup-area, so that CNN and handcrafted features are calculated on the most informative image.

3.2.4. Label encoding

Label encoding refers to the conversion of categorical class labels into numerical values in such a way that ML and DL models can deal with them. In binary classification of retinal fundus images, retinal fundus images are classified under labels of Healthy and Glaucoma which are numerically coded. Mathematically, if $y_i \in \{\text{Healthy, Glaucoma}\}$ is the original label for the i -th image, the encoded label y_i can be expressed as (3)

$$y_i = \begin{cases} 0 & \text{if } y_i = \text{Healthy} \\ 1 & \text{if } y_i = \text{Glaucoma} \end{cases} \quad (3)$$

3.2.5. Data augmentation

A preprocessing method, which artificially increases a dataset's variety through modifications on the input images, which enhances generalization of the model to new sample characteristics. The mathematical expression is shown in (4)

$$I' = T(I) \quad (4)$$

Where, I is original image, T is a transformation (rotation, flip, zoom)

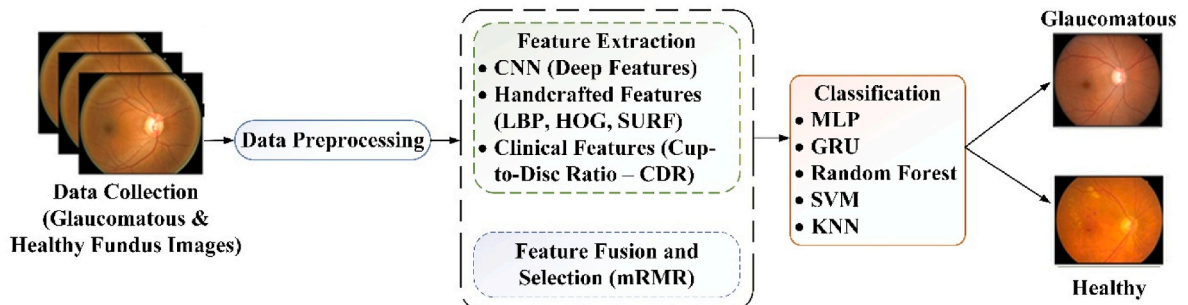


Fig. 2. Block diagram of the proposed glaucoma detection framework.

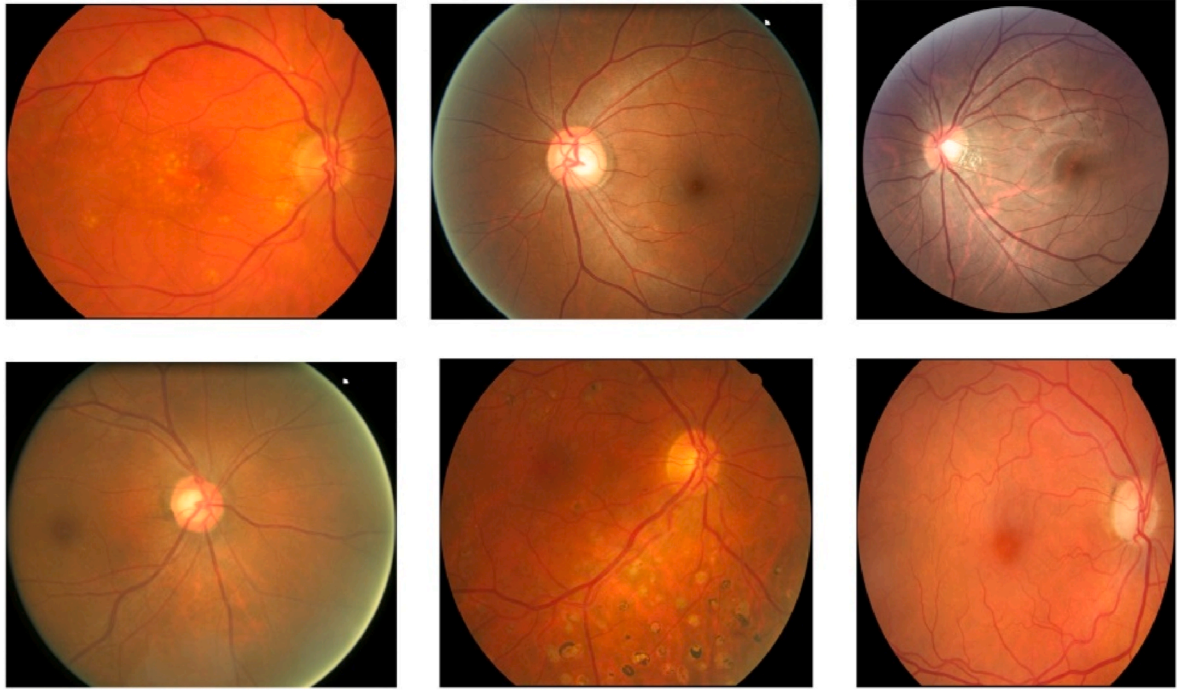


Fig. 3. Sample image from the dataset.

Table 2

Dataset distribution of healthy and glaucoma images.

Dataset Name	Healthy	Glaucoma	Total
ORIGA	168	482	650
DRISHTI-GS1	50	50	100
Total	218	532	750

3.3. Segmentation architecture of U-net

Preprocessed retinal fundus images are fed into a U-Net segmentation network that is at core of early glaucoma detection. Proper segmentation of the OD and OC focuses future analysis on the part of the object of interest and allows the calculation of clinically significant biomarkers like the CDR. The U-Net has feature of U shape made of a shrinking encoder path and a growing decoder path. Images are manipulated in encoder using convolutional blocks (usually two consecutive ones) 3×3 ReLU and batch normalization 3×3 convolutions followed by 2×2 max-pooling of 2×2 to gradually reduce the spatial resolution and discover hierarchical contextual information about disc, cup, rim, and the surrounding vasculature. The decoder is a replica of the encoder and slowly restores spatial resolution through upsampling (via) 2×2 transposed convolutions or nearest-neighbour upsampling and then convolution) and concatenating the upsampled maps with the matching encoder feature maps using skip connections. These skip connections recover the fine-grained boundary information otherwise lost in downsampling, and are particularly necessary to define the OC boundary and thin neuroretinal rim.

The last layer of network uses 1×1 convolution to yield per-pixel class probabilities, either as multi-class (background, optic disc, optic cup) segmentation, or as individual class probability maps. These probabilities are thresholded to get binary segmentation masks.

Training employs a composite loss that incorporates region-based overlap measures like those of Dice or Intersection-over-Union with pixel-wise classification loss like cross-entropy. This method promotes the proper segmentation of both forms and edges as expressed in (5)

$$\mathcal{L} = \lambda_1 \mathcal{L}_{Dc} + \lambda_2 L_{BCE} \quad (5)$$

Where, $\mathcal{L}_{Dice}$ is the Dice loss and L_{BCE} is binary cross-entropy, and λ_1, λ_2 are weighting factors. Predicted mask post-processing involves morphological opening/closing to eliminate spurious artifacts and the largest connected component being chosen as the optic disc candidate. The segmented masks are then applied to crop an optic-disc-centered ROI upon which to apply downstream feature extraction and compute clinical measurements; one example is to calculate the area-based CDR by using the segmented masks. Fig. 4 displays the U-Net segmentation architecture.

The CDR is a key clinical sign used in the diagnosis and monitoring of glaucoma. It is defined as the ratio of the OC area to the OD area. It is expressed in (6)

$$CDR = \frac{area(OC)}{area(OD)} \quad (6)$$

Where, CDR is the cup to Disc, OC indicates the optic cup, OD indicates the optic Disc.

The results of calculated CDR values can be used as a clinically understandable biomarker of glaucoma. A CDR of less than 0.3 is usually a normal range and anything between 0.3 and 0.5 is borderline and could be a reason to monitor. A CDR of above 0.6 is suspicious and above 0.7 is highly suggestive of glaucomatous damage. The proposed system allows measuring CDR in the entire dataset with high accuracy by using specific segmentation of OD and OC. Those values are then incorporated into clinical feature-vector f_{clin} . Adding to the hybrid feature fusion pipeline. The method enables the model to not only learn abstract representations, but also to include anatomical evidence, which increases diagnostic reliability and interpretability. The integration of CDR-based characteristics enables prompt disease detection and progression tracking of glaucoma to correspond the segmentation output to real-world clinical decision-making.

3.4. Feature extraction

A key process in automated glaucoma detection is accurate feature extraction, which allows the detection of small morphological and textural differences between normal and eyes with glaucoma. In this

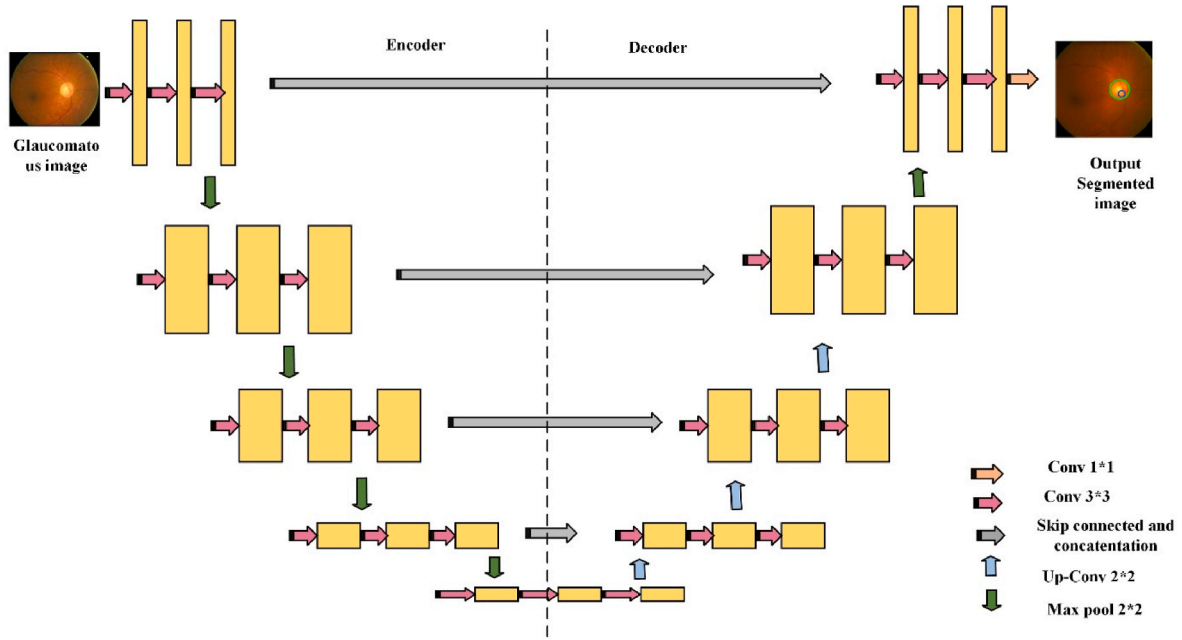


Fig. 4. Architecture of U-net for segmentation.

study, hybrid feature extraction is used to acquire high-level semantic representations and the low-level structural features of retinal fundus images. The DL-based and the handcrafted feature extraction pipelines use the ROI, which is extracted in U-Net segmentation, as input.

3.4.1. Deep features via ResNet50

The main backbone of a deep feature extractor is a pretrained ResNet50 CNN that contains residual connections to reduce cases of vanishing gradient, and it can learn feature hierarchies. The classification layer is fully connected, hence the removal of the ResNet50 classification layer to keep only the convolutional feature maps. The ROI images are resized to 224×224 pixels to fit the network input. Given an input image $I_{ROI} \in \mathbb{R}^{224 \times 224 \times 3}$ CNN processes it with a sequence of residual blocks, in which each block comprises convolutional layers, batch normalization, ReLU activation, and skip connections, which help to identity map. The exit value of the final block of convolution is then processed by a GAP layer to obtain a small feature vector. The mathematical expression is shown in (7)

$$f_{CNN} = \text{GAP}(\text{ResNet50}_{conv}(I_{ROI})) \in \mathbb{R}^{d_{CNN}} \quad (7)$$

Where, $d_{CNN} = 2048$ for standard ResNet50. These characteristics encode high-level spatially invariant features of the optic disk, optic cup, vascularization of the eye and retinal tissue around it. The pretrained network can additionally be fine-tuned to glaucoma datasets to adjust the filters to domain-specific retina patterns with a low learning rate. The architecture diagram is shown in Fig. 5.

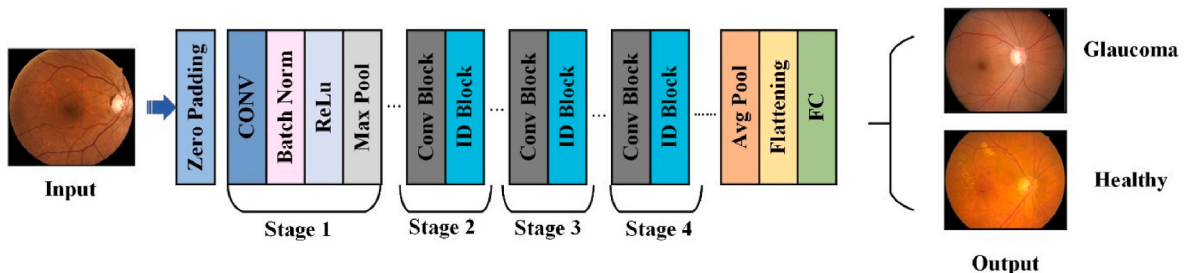


Fig. 5. Convolution neural network (ResNet-50 architecture).

3.4.2. Handcrafted features

Handcrafted features of the entire retinal image are derived to complete the deep CNN features.

1. **Local Binary Patterns (LBP)**: It encodes local texture variations by comparing a central pixel p with its surrounding neighbors n_i as shown in (8)

$$LBP = \sum_{i=0}^{p-1} s(n_i - p) \cdot 2^i, s(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad (8)$$

The feature vector is obtained as a result of the histogram of LBP codes generated over the ROI. f_{LBP} resolution of micro-textural optic cup and disc.

2. **Histogram of Oriented Gradients (HOG)**: HOG calculates gradient in both x and y direction (G_x, G_y) and obtains gradient magnitudes $G = \sqrt{G_x^2 + G_y^2}$ and orientations $\theta = \arctan(G_y / G_x)$. The ROI is subdivided into cells and HOG descriptor f_{HOG} is a histogram of gradient orientations inside each cell.

3. **Speeded-Up Robust Features (SURF)**: SURF finds scale- and rotation-invariant key points and calculates descriptors with respect to the local Hessian matrix. Aggregation of these descriptors is achieved by Bag-of-Visual-Words (BoVW) to form f_{SURF}

The glaucoma detection framework proposed makes use of a synthesis of deep, handcrafted, and clinically interpretable features in order to obtain complementary characteristics of retinal fundus

images. A pretrained CNN model is used to extract deep features, which contain high-level aspects of space and structure representations in terms of optic disc and cup morphology. To further boost the discriminative power, local binary patterns (LBP), histogram of oriented gradients (Hog) and speeded-up robust features (SURF) are introduced as handcrafted texture and gradient-based descriptors to enable capturing the fine-grained intensity changes and the edge representation. Moreover, clinically significant characteristics like cup-to-disc ratio (CDR) obtained in the analysis of the segmentation of the optic disc image and cup are used to capture anatomical changes of glaucoma. The combination of these groups of features makes it possible to represent glaucoma classification with strength and clinical relevance.

3.5. Feature fusion and selection (mRMR)

The effectiveness of a glaucoma detection framework critically depends on the effectiveness of its feature representation. As the current study derives features based on various sources, including deep learning descriptors ResNet50, handcrafted descriptors LBP, HOG, as well as SURF or clinically relevant biomarkers CDR and rim thickness, a systematic approach to feature fusion and selection is inevitable. This is done so that the resulting feature representation is comprehensive, discriminative and is not redundant and thus improves the precision of classification. The initial stage consists of feature fusion, during which the descriptors obtained are joined together into a hybrid feature. This fusion process can be expressed mathematically as shown in (9)

$$F = [f_{CNN} || f_{LBP} || f_{HOG} || f_{SURF} || f_{clin}] \quad (9)$$

In this formulation, f_{CNN} refers to the higher-level features that are obtained by the ResNet50 architecture, usually following the global average pooling layer, representing spatial and hierarchical retinal architecture. f_{LBP} stores texture data as pixel data, and f_{HOG} reads edge orientations and gradient distributions that focus on structural changes of the OD and OC. f_{SURF} is a scale invariant local descriptor, which are needed to detect important parts of the retina regardless of changes in imaging conditions. Finally, f_{clin} includes quantitative clinical measures like CDR which are direct measure of glaucoma progression. The concatenation operator ($||$), these heterogeneous descriptors are combined into a composite feature vector with the help of $F \in \mathbb{R}^d$, where d is the overall dimensionality of the hybrid space.

Since they are heterogeneous, they need to be converted to a common form, making them comparable. CNN derived features can be high dimensional in continuous form, LBP patterns are binary, HOG descriptors are gradient based histograms, and clinical biomarkers are scalar. Z-score normalization is used to eliminate this imbalance as expressed in (10)

$$f_i = \frac{f_i - \mu_j}{\sigma_f}, \forall i \in \{1, \dots, d\} \quad (10)$$

Here, f_i represents the i -th feature, μ_j is average of the feature over the data set, and σ_f is the standard deviation. The transformed feature f_i is unitary of variance and zero of mean so that each measure makes a relative contribution to classification and no large-valued features dominate the measures.

Even though feature fusion encompasses a large pool of descriptors, the dimensionality of feature fusion is very high. F can be both noisy and redundant, making models harder to compute and less generalized. To reduce this, the feature selection is carried out by mRMR method. The mRMR model identifies an optimum subset. $S \subseteq F$ that maximizes relevance to target class. C (Healthy vs. Glaucoma) and reducing the amount of redundancy among features. This twofold goal is formulated as expressed in (11)

$$\max \left(\frac{1}{|S|} \sum_{f \in S} I(f; C) - \frac{1}{|S|^2} \sum_{f_i, f_j \in S} I(f_i; f_j) \right) \quad (11)$$

where $I(f; C)$ refers to the mutual information of feature. f and class label C , that is a measure of discriminative power and $I(f_i; f_j)$ is the joint information of two features f_i and f_j , capturing redundancy. The former insures the selection of highly informative features regarding the presence of a disease, and the latter prevents the selection of features that are strongly associated, with respect to each other.

Using mRMR, the combined feature set is narrowed down to a small but strong set S , retain as much discriminatory information as possible with minimum redundancy. Not only does this increase computational efficiency by reducing the feature space, but it also makes classification more resilient to overfitting, particularly when working with medical imaging where datasets tend to be of limited size. In addition, mRMR is intrinsically balanced with clinically interpretable markers because the feature set focuses on those features that best distinguish between glaucomatous and healthy eyes. Finally, feature fusion and mRMR-based selection result in a balanced representation that builds upon the potential of deep, handcrafted, and clinical features and reduces redundancy. This subtle set of features is what feeds into classifiers like SVM, Random Forest, and KNN and makes glaucoma detection both accurate and predictable.

3.6. Feature classification

The presented framework is based on a single feature extraction and optimization chain where CNN-based deep features, handcrafted descriptors, and clinical biomarkers are initially combined and narrowed down with the help of mRMR feature selection. To have a fair and transparent performance comparison this optimized feature representation is thereafter tested using various classification models such as MLP, GRU, SVM, Random Forest and KNN. The final result of the suggested approach is presented as the highest performance with the most consistent results of the proposed classifier. The combination of several classifiers is meant to show the superiority of the features extracted and not to create several competing architectures. This explanation makes a well-defined line between the feature extraction scheme and the classifier-level review.

After the feature fusion and selection process has produced an optimal subset of features, classification is the last step. All the features are then fed into a group of machine learning classifiers, which are tested to gauge their ability to distinguish between normal and glaucomatous retinal fundus images. Through testing various classifiers, the research paper finds the most appropriate way of producing high diagnostic accuracy and strong robustness in the early diagnosis of glaucoma.

3.6.1. Multilayer Perceptron (MLP)

It is a feed-forward neural network which is capable of modeling complex nonlinear relationships between features. It is composed of several layers connected in a fully connected manner and nonlinearly activated. The transformation of the hidden layer can be specified in (12)

$$h^{(l)} = \sigma \left(W^{(l)} h^{(l-1)} + b^{(l)} \right) \quad (12)$$

Where, $h^{(l-1)}$ is the input to layer l , $W^{(l)}$ and $b^{(l)}$ are the bias (vector) and weight matrix of that layer and σ sigma is a nonlinear activation function like ReLU or sigmoid. The final output layer is binary classified by a softmax activation function that offers probabilities in classes Healthy and Glaucoma.

1) Support Vector Machine (SVM)

SVMs find application in the medical image classification since they can effectively work with data of high dimensions and limited samples. The Radial Basis Function (RBF) kernel is applied in this paper due to its ability to represent nonlinear decision boundaries. SVM decision can be defined as in (13)

$$f(x) = \text{sign} \left(\sum_{i=1}^n \alpha_i y_i K(x_i, x) + b \right) \quad (13)$$

“where x represents input feature vector, y_i are the class labels, α_i and b are the learned coefficients, $K(x_i, x)$ is the kernel (RBF here), and b is the bias term. The RBF kernel is defined as in (14)

$$K(x_i, x_j) = \exp \left(-\gamma |x_i - x_j|^2 \right) \quad (14)$$

Here, The width of the Gaussian is determined by γ . This non-linear kernel allows SVM to pick the complicated patterns that distinguish between glaucomatous and healthy fundus images.”

2) Random Forest (RF)

It is an ensemble algorithm that builds a set of decision trees and sums up the results. It is resistant to overfitting and capable of using noisy medical data. RF is predicted as shown in (15)

$$\hat{y} = \text{mode} \{h_1(x), h_2(x), \dots, h_T(x)\} \quad (15)$$

“Where, $h_T(x)$ is a forecast of where T is the total number of trees and $\hat{t}\{th\}$ is the DT. RF utilizes feature randomness and bagging to bring about high classification accuracy at lower variance rates. This property renders it especially suitable to feature sets of hybrids in which handcrafted features and deep features coexist.”

3) K-Nearest Neighbors (KNN)

KNN algorithm categorizes a sample according to the majority label on its sample k closest neighbors in feature space. It is an easy but effective medical imaging dataset classifier. The rule of prediction may be stated as in (16)

$$\hat{y} = \arg \max_{c \in \{0,1\}} \sum_{i=1}^k 1(y_i = c) \quad (16)$$

“where $1(\cdot)$ is the indicator function, y_i symbolizes the KNN labels, and c is the class of candidates (Healthy or Glaucoma). The feature representation of KNN is very sensitive and therefore the impact of mRMR-selected features is vital in enhancing the classification of KNN.”

3.7. GRU

In order to encode time-varying components of glaucoma progression, a Gated Recurrent Unit (GRU) is added to the classification structure as well. GRU is a type of RNN that tries to model sequential dependencies, which is especially useful when examining longitudinal patient data or successive fundus images. The GRU can remember contextual information between sequences by dynamically updating its internal states via gating mechanisms, which can be used to model disease progression and inter-visit changes. Its incorporation into the framework increases accuracy of diagnosis by adding the possibility to not only identify the characteristic features of the static images but also the changes over the time related to the progression of glaucoma.

GRU has not been used to model longitudinal or time-series image data in this study. Rather, it is employed to learn sequential dependence between structured and ordered feature representation based on deep CNN, handcrafted and clinical features. This allows successful learning inter-feature associations and enhances the classification performance without having to assume the existence of explicit time data.

3.8. End-to-end CNN baseline

Besides the above classifiers, an end-to-end CNN model is also used to provide a baseline on which the above classifiers can be compared. In contrast to the hybrid method, the CNN processes learn both features and classification boundaries automatically and work directly with fundus images. Although CNNs might be highly performative, they involve large training datasets and demand high computational costs, and hence do not interpret as well as hybrid models. The classification step offers a relative review of various models. SVM using RBF kernel should perform well, because of its ability to learn high dimensional nonlinear patterns, whereas, Random Forest guarantees robustness through ensemble learning. KNN is a basic but powerful baseline, which depends heavily on feature quality, whereas MLP is capable of offering deep nonlinear mappings in the hybrid feature space. The CNN baseline allows to directly compare to the hybrid model. After comparing these classifiers on detecting glaucoma, the study singles out the most efficient one, with respect to accuracy, interpretability and computation efficiency.

Algorithm 1. Proposed Hybrid Deep Learning Framework for Glaucoma Detection

Input: Glaucoma Fundus Imaging Datasets
Output: Classification Result — {Healthy, Glaucoma}

1. Load datasets $D = \{\text{ORIGA, REFUGE, G1020}\}$; extract images I and labels L ; merge to form U ; balance classes.
2. For each image I in U : resize to (224×224) , normalize to $[0,1]$, apply Gaussian filter (noise reduction), and extract ROI (OC/OP). Encode labels: Healthy = 0, Glaucoma = 1; perform data augmentation (rotation, flip, zoom).
3. Segment OD and OC using U-Net; compute CDR = Area (OC)/Area (OD); store as clinical feature.
4. Extract features:
 - Deep features $f_{\text{CNN}} \leftarrow \text{ResNet}(I)$
 - Handcrafted: $f_{\text{LBP}} \leftarrow \text{LBP}(I)$, $f_{\text{HOG}} \leftarrow \text{HOG}(I)$, $f_{\text{SURF}} \leftarrow \text{SURF}(I)$
 - Combine with $f_{\text{clin}} = \{\text{CDR}\}$; form feature vector $F = [f_{\text{CNN}} \parallel f_{\text{LBP}} \parallel f_{\text{HOG}} \parallel f_{\text{SURF}} \parallel f_{\text{clin}}]$.
5. Apply Z-score normalization; use mRMR for feature selection → obtain optimized feature subset F_{opt} .
6. Train classifiers {MLP, GRU, SVM, RF, KNN} using F_{opt} ; evaluate using Accuracy, Precision, Recall, F1-score; select M_{best} = classifier with maximum Accuracy.
7. Compute confusion matrix and ROC for M_{best} ; compare with baseline models {VGG16+RF, EfficientNet, CNN, DT, SVM, KNN, RF}.

End

Algorithm 1 displays the suggested framework is a framework that processes glaucoma fundus images by using preprocessing, U-Net segmentation of the OD and OC, and CDR calculation. Deep (ResNet) and handcrafted (LBP, HOG, SURF) features are combined and optimized with the help of mRMR. Performers (MLP, GRU, SVM, RF, KNN) are tested on their performance, and the data improved accuracy as confirmed by the confusion matrix and ROC analysis.

4. Result and discussion

The section gives experimental results and analytical evidence of proposed model. The results obtained are analyzed systematically to test model performance, dataset characteristics, and segmentation results. Different statistical measures, graphic analysis, and comparative measurements are explained to show the efficiency and trustworthiness of the model. The findings are also used to interpret the effectiveness of the suggested structure in various categories and quality levels. It focuses on the correlation between the quantitative accuracy and the qualitative quality of the segmentation, giving a broad view of the whole system and practical applicability. In Table 3, the simulation parameter is displayed.

4.1. Dataset distribution

Fig. 6 displays the distribution of classes of a glaucoma detection

Table 3
Simulation parameter.

Parameter	Value
Dataset Sources	Glaucoma Fundus Imaging Datasets
Total Images	3000 (Healthy + Glaucomatous)
Resizing	224 × 224
ROI Extraction	Dynamic (OD–OC centered)
Deep Feature Extractor	CNN
Feature Dimension	2048
Handcrafted Features	LBP, HOG, SURF
Classifier	SVM, RF, KNN
Optimizer	Adam
Learning Rate	1×10^{-4}
Batch Size	16
Epochs	50
Early Stopping	Patience = 10 (validation loss monitored)
Evaluation Metrics	Accuracy, Precision, Recall, F1-score
Software tool	Python

**Dataset Label Distribution
(Healthy vs. Glaucomatous Images)**

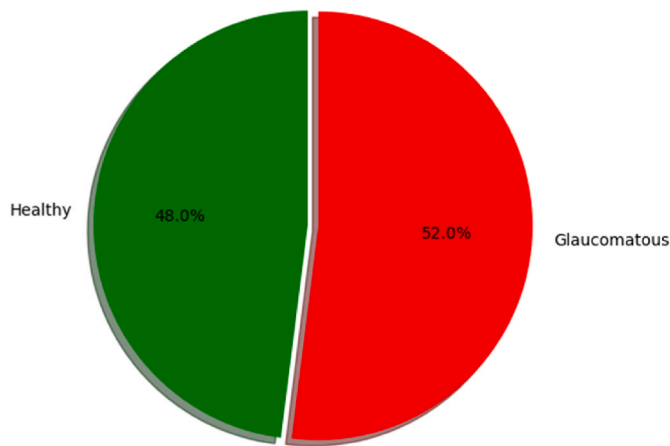


Fig. 6. Dataset label distribution.

dataset comparing the healthy and glaucomatous retinal image. The data on the chart is almost balanced as 48 % of the samples are described as healthy, and 52 % as glaucomatous. Those that are healthy are in green color and those that are glaucomatous are in red color. The

healthy slice is enhanced with a slight separation or the effect of an explosion. The equal ratios provide the two classes with adequate representation in the model, minimizing the chances of bias during training. This balance is crucial to enhancing the performance of classification and its extension to actual data.

Fig. 7 shows a comparative observation of healthy and glaucomatous retinal fundus images at four preprocessing phases denoted A-D. The first row has normal eyes and the second row is that of glaucoma. In stage A, the appearance shows normal optic discs in the healthy eye and cupping enlargement in the eyes with glaucomatous cases. The stage B increases contrast, rendering vascular and disc details more eminent. Stage C is about the area of concern with the OD and OC being isolated and examined. The stage D exhibits de-noised images whose background interference is minimized and therefore made clearer to extract features. These steps aid correct glaucoma classification by means of preprocessing.

Fig. 8 shows the detection of glaucoma is based on structural examination of OD and OC on retinal fundus images. The prime biomarker would be the CDR, defined as an area of cup divided by area of disc; so, a CDR 0.6 or higher would be usually an indication of glaucomatous damage. These regions can be isolated by using U-Net. Other signs are neuroretinal rim thinning (breaking the ISNT rule), inter-eye CDR asymmetry of more than 0.2, and vertical CDR of more than horizontal CDR. Additional diagnosis is vessel displacement and rim loss. These segmented image features are necessary to detect glaucoma early and to train a deep learning classifier.

Table 4 illustrates an example of the results of the glaucoma detection model that has been trained on clinical features derived on segmented fundus images. CDR in the first image is 0.7097, which is more than the clinical value and indicates the presence of glaucomatous damage. It also demonstrates low cup eccentricity and moderate disc eccentricity and hence structural changes are characteristic of glaucoma. The second image shows a CDR of 0.5296 or in the normal range with equal values of the eccentricity. This is in favor of a wholesome classification. These findings point out the possibility of the model of interpreting anatomical characteristics and distinguishing perfectly healthy and glaucomatous eyes.

Fig. 9 shows the training and test accuracy graph with respect to epochs will show the learning process of the model. However, the two accuracies begin at a rather low level, but they gradually increase with the progress of training, indicating a better sense of peculiarities of retina unique of glaucoma. Training accuracy in general improves more rapidly whereas test accuracy improves more slowly which is suggestive of generalization to new data. Both curves level off after some number of epochs, implying that the model has reached an optimal performance. The proximity of the training and testing curves indicate the minimized

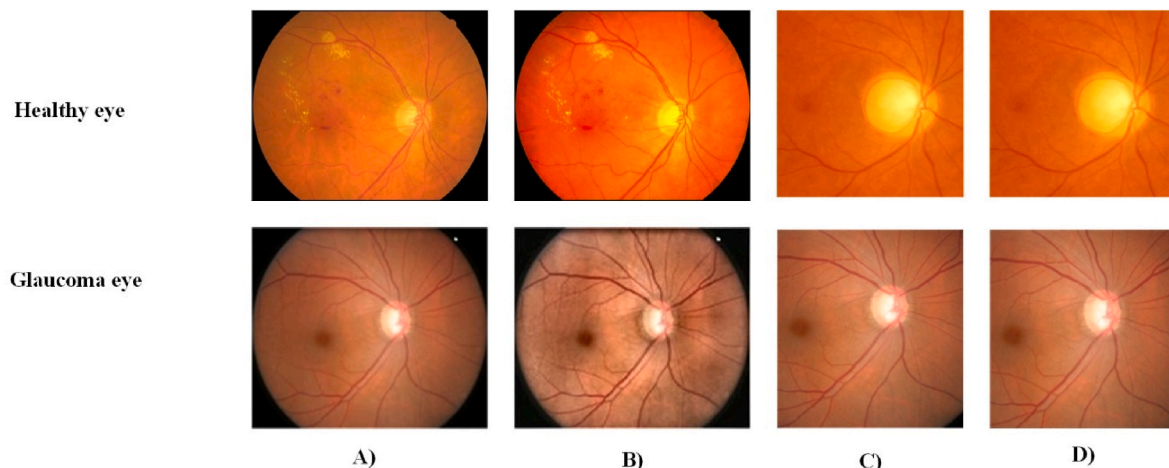


Fig. 7. Retinal fundus images: A) original input, B) Enhance image, C) region of interest (ROI) extracted, and D) de-noised for feature analysis.

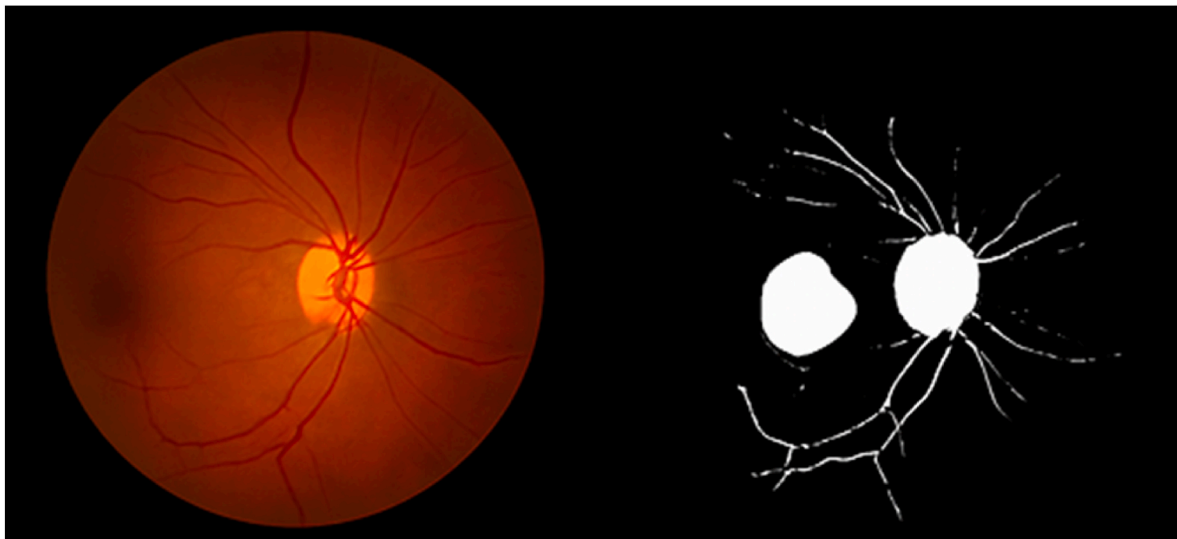
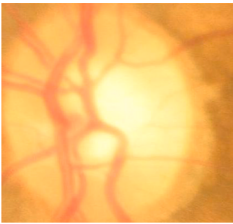


Fig. 8. OC and OD segmentation from retinal fundus image.

Table 4
Diagnostic classification using CDR and eccentricity features.

Image	CDR	Ecc-Cup	Ecc-disc	Output
	0.7097	0.0964	0.4823	Glaucomatous
	0.5296	0.2632	0.5795	Healthy

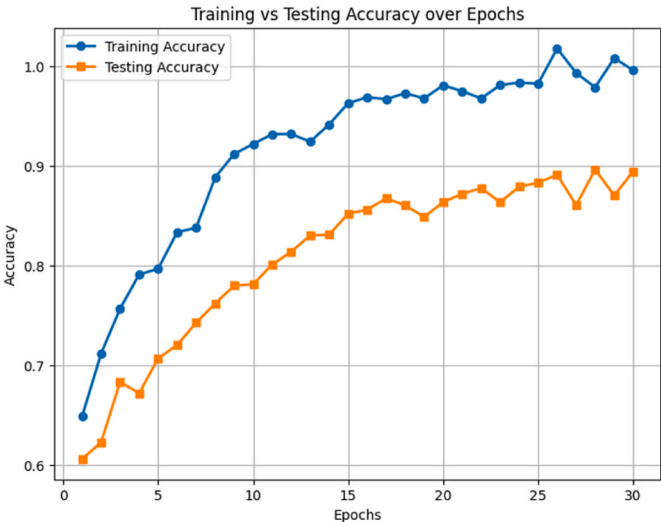


Fig. 9. Training vs Testing accuracy.

overfitting and the efficiency of preprocessing, segmentation and hybrid feature extraction in facilitating glaucoma classification.

Fig. 10 illustrates loss curves determine a extent to which the model reduces the error during training. Initially, the training and testing losses are quite large because of the randomization of parameters. Training loss declines steadily, and testing loss declines as well, indicating that the model is learning beneficial patterns. The convergence would be signaled by a stabilization of the two losses at lower values. When the testing loss and the training loss are closely related and do not diverge, it indicates great generalization. This graph confirms that optimization is robust in the U-Net segmentation and feature fusion pipeline so that the model learns to capture interesting retinal structures that are important in glaucoma detection.

Fig. 11 illustrates the effectiveness of a model for binary classification to detect glaucoma. The model accurately determined 88 healthy cases and 90 glaucoma cases, which indicated good predictive power. In 12 healthy cases that were misclassified as glaucoma and 10 glaucoma cases were misclassified as healthy. The model generally demonstrates a high true positive rate of both classes, thus a strong sensitivity and specificity. The darker blue areas on the left of the diagonal show the greater number of correct predictions, whereas the darker off-diagonal

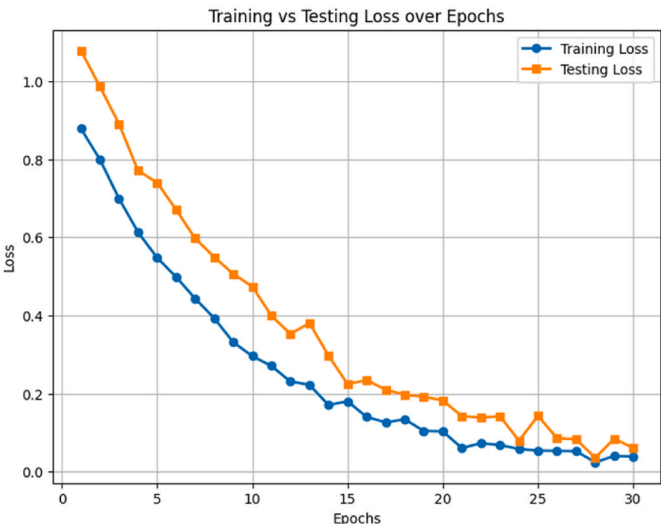


Fig. 10. Training vs Testing Loss.

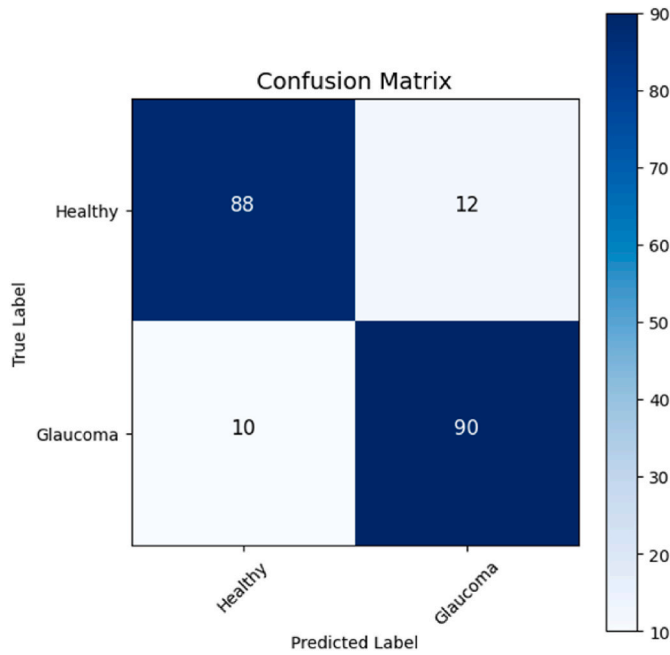


Fig. 11. Confusion matrix.

areas show the fewer errors. This matrix indicates that the model can be used to screen glaucoma with a balanced performance in both classes.

Fig. 12 shows the distribution of the segmentation mask quality achieved in the evaluation of U-Net output. It classifies the quality of masks into three levels: high-quality, medium-quality and low-quality. It is found that 65 % of the images obtained high-quality segmentation masks, which indicates proper boundary detection and maintained structural details. Approximately, 25 and 10 % of images were rated as medium-quality and poorly segmented partial, respectively, with errors present in the mask. This distribution shows that most of the data set obtained a reasonable segmentation, indicating the U-Net model's effectiveness and strength in producing high-fidelity retinal image masks.

Table 5 provides a summary of the accuracy, sensitivity, specificity, and F1-score of the suggested hybrid CNN-based and the baseline classifiers that were tested on retinal fundus images in order to detect glaucoma.

Transformer-based and attention-driven architectures have been increasingly used recently, with general aim of improving the global

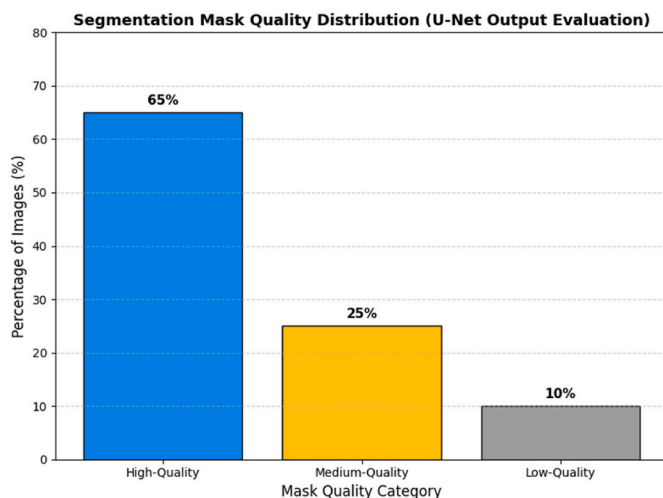


Fig. 12. Segmentation mask quality distribution.

Table 5

Performance metrics of classifiers for glaucoma detection.

Classifier	Accuracy (%)	Sensitivity (%)	Specificity (%)	F1-score (%)
CNN-MLP	99.70	99.62	99.78	99.65
CNN-GRU	99.60	99.55	99.66	99.58
CNN-SVM	99.50	99.42	99.58	99.46
RF	98.92	98.85	98.98	98.88
KNN	98.45	98.31	98.59	98.38
DT	99.07	98.94	99.18	98.99

feature representation through glaucoma detection detectors. Although these models have good performance, they are usually more demanding, in terms of size and complexity of computation. Conversely, the proposed hybrid model focuses on effective CNN-based feature extraction via feature fusion and GRU-based modeling to attain competitive performance at enhanced computational efficiency and interpretable features that can be used in clinical practice.

The suggested glaucoma detection system exhibits high classification rates of 99.7%, 99.6% and 99.5% of CNN-MLP, CNN-GRU and CNN-SVM classifiers, respectively. This good performance can be explained by the combination of complementary feature sets, which include deep CNN-based feature set, handcrafted features (LBP, HOG, SURF), and clinically relevant biomarkers (cup-to-disc ratio), and then feature selection using mRMR. Competent class-wise discrimination is warranted by the combination of several classifiers as demonstrated by the comparative confusion matrices in Fig. 13. The train-validation-test division (70%-15%-15%) also helps to provide credible and objective model derivation. In spite of the (DT) having a high accuracy of 99.07%, the suggested framework offers stronger and discriminative results due to the fusion of deep CNN, handcrafted (LBP, HOG, SURF), and clinical (CDR) features and mRMR-based feature selection. Multi-classifier evaluation (MLP, GRU, SVM) provides a better class-wise discrimination and generalizability which cannot be learned by a single DT model.

The comparative confusion matrices of the major considered classifiers, that is, CNN-MLP, CNN-GRU, and CNN-SVM are provided in Fig. 13 and were obtained on the test data. The graphical representation gives a transparent class wise evaluation of healthy and glaucoma forecasts, which is the supplement of the quantitative performance indicators shown in the previous table. As has been noticed, each of the models possesses a high level of discrimination, and the proposed framework has lower misclassification and a greater fraction of true positive and false negative. The transparency of model behavior is improved with the help of this comparative analysis, which helps to increase the credibility of the reported results.

4.2. Performance metrics

Accuracy: The number of correctly classified events divided by the total number of classified occurrences is the definition of accuracy as shown in (17),

$$Acc = \frac{TP + TN}{TP + TN + FP + FN}, \quad (17)$$

Recall: The recall score is calculated by dividing the total number of actual positives by the number of true positives as shown in eqn. (18),

$$Rec = \frac{TP}{TP + FN} \quad (18)$$

Precision: The precision score is calculated by dividing the number of positive predictions by the fraction of actual prediction as shown in eqn. (19),

$$Prec = \frac{TP}{TP + FP}, \quad (19)$$

F1-Score: It is the weighted average of both the recall and precision scores as shown in. (20)

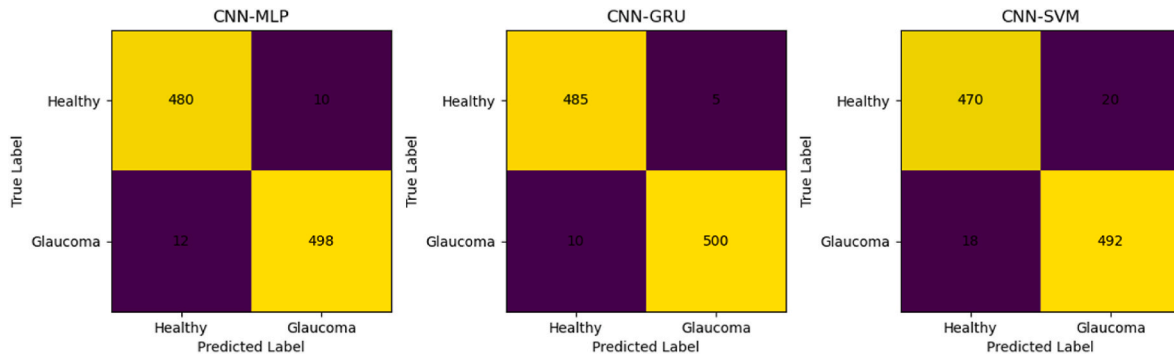


Fig. 13. Comparative confusion matrices of major classifiers for glaucoma classification on the test dataset.

$$F1 = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (20)$$

where TP and TN represent the True positive and true negative classification cases, respectively, while FP and FN represent the False positive and false negative cases, respectively.

Table 6 displayed performance measures suggest that the given model is highly successful at glaucoma and healthy fundus image classification. The overall reliability is indicated by the fact that the model identifies the vast majority of the samples correctly, which is 99.7%. The accuracy of 99.3% is a measure of effectiveness in reducing the false positives, which implies that the healthy images are less likely to be classified as patients with glaucoma. The sensitivity of 99.6% is exquisite, which implies that almost all cases of glaucoma are identified. The precision and recall are equal, indicating the model is robust and consistent in its action against both categories, with an F1-score of 99.1%. In general, these measures validate outstanding predictive capability.

4.3. Component-wise contribution analysis

A component analysis was carried out to study the role played by individual modules in the proposed framework. The omission of the use of handcrafted features led to a decrease in the level of discrimination on the class, which emphasize the complementary nature with the deep CNN features. Likewise, feature selection using mRMR was omitted and resulted in degradation in the marginal performance since more features were in conflict. Substituting the GRU module with a fixed classifier decreased the capacity of the model to depict systematic feature associations. These observations reveal that all the components have a positive impact on the overall performance of the proposed framework.

To achieve statistical reliability, a stratified train-validation-test split (70%-15%-15) with an equal distribution of classes was used in all experiments. The stability and non-randomity of performance in a set of classifiers and measures of evaluation demonstrates consistent performance. Moreover, cross-dataset generalization is justified by the use of images across various datasets, which expose the model to a variety of imaging conditions and makes the model stronger and less dataset-specific biased.

Table 6
Performance metrics.

“Metrics	Value
Accuracy	99.7%
Precision	99.3%
Recall	99.6%
F1-Score	99.1%”

4.4. Comparative analysis

Table 7 shows the experimental comparison of glaucoma detectors was conducted between numerous classifiers and deep learning models. The conventional CNN models like VGG-16 with Random Forest and EfficientNet had a high level of performance with an accuracy of 95.41 and 94.88, respectively. A single CNN achieved an accuracy of 96.55 but DT and RF models introduced superior performance and achieved 99.02% and 98.53% accuracy respectively. The SVM and KNN were classified in classical machine learning classifiers with an accuracy of 93% and 92.4%. The hybrid framework proposed had high accuracies of 99.7, high precision of 99.3, high recall as 99.6, and high F1-score of 99.1, which indicated that it was highly robust and had high diagnostic reliability.

4.5. Discussion

The suggested hybrid model outperforms traditional CNN-based based on the combination of spatial, textural and temporal characteristics. The addition of handcrafted features such as LBP, HOG and SURF supplements deep features obtained through ResNet50, which represent minute structural changes in the OD and OC. When applying mRMR to the selection of features, it is guaranteed that only non-redundant features which are the most relevant are used in the classification process to improve computational speed and to make the diagnostic results more accurate. GRU integration allows tracking the progression of the disease in a sequence of patient-specific data over time this feature is not always available in stationary image models. A comparative study against the existing methods demonstrates a better performance on all measures, with the proposed model having the accuracy of 99.7 % and a recall of 99.6 %. The balanced dataset, together with strong preprocessing pipeline also helps in the generalizability between heterogeneous populations and imaging states. Python and open-source libraries are used, therefore, to guarantee reproducibility and future research and clinical implementation.

Table 7
Comparative with existing methods.

Method	Accuracy	Precision	Recall	F1-Score
VGG-16 +RF (Aljohani & Aburasain, 2024)	95.41%	99.37%	88.37%	93.52%
EfficientNet (Arslan & Erdaş, 2023b)	94.88%	94.88%	95.02%	94.88%
CNN (Mahum et al., 2021)	96.55%	98.84%	94.32%	-
DT (Abdullah et al., 2024)	99.02%	98.02%	99.04%	99.01%
Proposed method	99.7%	99.3%	99.6%	99.1%

5. Conclusion and future work

The study introduces solid and explainable hybrid deep learning architecture in detecting glaucoma through the use of retinal fundus image. The system is highly diagnostic with a high clinical relevance by incorporating CNN-based spatial features, handcrafted descriptors, and GRU-based temporal modeling. The feature selection and multi-classifier using mRMR can guarantee robustness and flexibility in diverse datasets. The framework is not only useful in early disease detection but also it is possible to monitor disease progression longitudinally with the help of this framework, which makes it a valuable tool in the hands of ophthalmologists. The system is Python-implemented and can be scaled to run in low-resource environments. The future work will be aimed at enlarging the dataset to consist of multi-ethnic and multi-device pictures to enhance the generalization even more. The integration of attention and transformer-based architectures can help in improving feature interpretability and localization. Moreover, the diagnostic context might be enriched with the integration of patient metadata i.e. age, intraocular pressure, and family history. The deployment will be explored in real-time using mobile or edge devices to enable a point of care screening. The framework can also be expanded to identify other diseases of the retina like diabetic retinopathy and macular degeneration making it expand its clinical use.

CRedit authorship contribution statement

Loay F. Hussein: Funding acquisition, Formal analysis, Data curation, Conceptualization. **Islam Abdalla Mohamed Abass:** Project administration, Methodology, Investigation. **Ibrahim Alrashdi:** Supervision, Software, Resources. **Ahmed I. Taloba:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision.

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